Lecture Notes: Chapter 4

Sections 4.1, 4.2, 4.4, 4.5, 4.6, 4.7, 4.9

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Section 4.1 - Maxima and Minima

With a working understanding of derivatives, we now undertake one of the fundamental tasks of calculus: analyzing the behavior and producing accurate graphs of functions. An important question associated with any function concerns its maximum and minimum values: On a given interval (perhaps the entire domain), where does the function assume its largest and smallest values? Questions about maximum and minimum values take on added significance when a function represents a practical quantity, such as the profits of a company, the surface area of a container, or the speed of a space vehicle.

> **Definition 1** (Absolute Maximum and Minimum).
> Let f be defined on a set D containing c. If $f(c) \geq f(x)$ for every x in D, then $f(c)$ is an absolute maximum value of f on D. If $f(c) \leq f(x)$ for every x in D, then $f(c)$ is an absolute minimum value of f on D. An absolute extreme value is either an absolute maximum or an absolute minimum value.

Notice: The existence and location of absolute extreme values depend on both the *function* **and** the *interval* of interest.

> **Theorem 1** (Extreme Value Theorem).
> *A function that is continuous on a closed interval $[a,b]$ has an absolute maximum value and an absolute minimum value on that interval.*

> **Definition 2** (Local Maximum and Minimum Values).
> Suppose c is an interior point of some interval I on which f is defined. If $f(c) \geq f(x)$ for every x in I, then $f(c)$ is a local maximum value of f. If $f(c) \leq f(x)$ for every x in I, then $f(c)$ is a local minimum value of f.

In the textbook, the authors adopt the convention that local maximum values and local minimum values occur only at interior points of the interval(s) of interest. We will also adopt this convention for the sake of consistency.

> **Theorem 2** (Local Extreme Value Theorem).
> *If f has a local maximum or minimum value at c and $f'(c)$ exists, then $f'(c) = 0$.*

Note: Local extrema can also occur at points c where $f'(c)$ does not exist. Because local extrema may occur at points c where $f'(c) = 0$ and where $f'(c)$ does not exist, we make the following definition:

> **Definition 3** (Critical Point).
> An interior point c of the domain of f at which $f'(c) = 0$ or $f'(c)$ fails to exist is called a critical point of f.

Note: The converse to the Local Extreme Value Theorem is false. Namely, it is possible that $f'(c) = 0$ at a point without a local maximum or local minimum value occurring there (see, for example, $f(x) = x^3$). Therefore, critical points are candidates for the location of local extreme values, but you must determine whether they actually correspond to local maxima or minima.

Example 1. Find the critical points of $f(x) = x^2 ln(x)$.

Note that f is differentiable on its domain, which is $(0, \infty)$. By the Product Rule,

$$f'(x) = 2xln(x) + x^2 \frac{1}{x} = x\big(2ln(x) + 1\big).$$

Setting $f'(x) = 0$ gives $x(2ln(x)+1) = 0$, which has the solution $x = e^{-1/2} = 1/\sqrt{e}$. Because $x = 0$ is not in the domain of f, it is not a critical point. Therefore, the only critical point is $x = 1/\sqrt{e} \approx 0.61$. A graph of f (See Figure 4.10 in your textbook) reveals that a local (and, indeed, absolute) minimum value occurs at $1/\sqrt{e}$, where the value of the function is $f(1/\sqrt{e}) = -1/(2e)$.

Theorem 1 guarantees the existence of absolute extreme values of a continuous function on a closed interval $[a, b]$, but it doesn't say where these values are located. To determine this, we use the following procedure:

Proceedure

Assume the function f is continuous on the closed interval $[a, b]$.

1. Locate the critical points c in $[a, b]$, where $f'(c) = 0$ or $f'(c)$ does not exist. These points are candidates for absolute maxima and minima.
2. Evaluate f at the critical points and at the endpoints of $[a, b]$.
3. Choose the largest and smallest valued of f from step 2 for the absolute maximum and minimum values, respectively.

Example 2. Find the absolute maximum and minimum values of $f(x) = x^4 - 2x^3$ on the interval $[-2, 2]$.

Because f is a polynomial, its derivative exists everywhere. So if f has critical points, they are points at which $f'(x) = 0$. Computing f' and setting it equal to zero, we have

$$f'(x) = 4x^3 - 6x^2 = 2x^2(2x - 3) = 0.$$

Solving this equation gives the critical points $x = 0$ and $x = 3/2$, both of which lie in the interval $[-2, 2]$; these points and the endpoints are candidates for the location of absolute extrema. Evaluating f at each of these points, we have

$$f(-2) = 32, \quad f(0) = 0, \quad f(3/2) = -27/16, \quad \text{and} \quad f(2) = 0.$$

The largest of these function values is $f(-2) = 32$, which is the absolute maximum of f on $[-2, 2]$. The smallest of these values is $f(3/2) = -27/16$, which is the absolute minimum of f on $[-2, 2]$. A graph of f (see figure 4.11 in your textbook) shows that the critical point $x = 0$ corresponds to neither a local maximum nor a local minimum.

Example 3. Find the absolute maximum and minimum values of $f(x) = x^{2/3}(2-x)$ on the interval $[-1,2]$.

Differentiating $g(x) = x^{2/3}(2-x) = 2x^{2/3} - x^{5/3}$, we have

$$g'(x) = \frac{4}{3}x^{-1/3} - \frac{5}{3}x^{2/3}.$$

Because $g'(0)$ is undefined and 0 is in the domain of g, $x = 0$ is a critical point. In addition, $g'(x) = 0$ when $4 - 5x = 0$, so $x = 4/5$ is also a critical point. These two critical points and the endpoints are candidates for the location of absolute extrema. The next step is to evaluate g at the critical points and endpoints:

$$g(-1) = 3, \quad g(0) = 0, \quad g(4/5) \approx 1.03, \quad \text{and} \quad g(2) = 0.$$

The largest of these function values is $g(-1) = 3$, which is the absolute maximum value of g on $[-1,2]$. The least of these values is 0, which occurs twice. Therefore, g has its absolute minimum value on $[-1,2]$ at the critical point $x = 0$ and the endpoint $x = 2$.

Example 4. A stone is launched vertically upward from a bridge 80 ft above the ground at a speed of 64 ft/s. Its height above the ground t seconds after the launch is given by

$$f(t) = -16t^2 + 64t + 80, \quad \text{for} \quad 0 \le t \le 5.$$

When does the stone reach its maximum height?

We must evaluate the height function at the critical points and at the endpoints. The critical points satisfy the equation

$$f'(t) = -32t + 64 = -32(t-2) = 0,$$

so the only critical point is $t = 2$. We now evaluate f at the endpoints and at the critical point:

$$f(0) = 80, \quad f(2) = 144, \quad \text{and} \quad f(5) = 0.$$

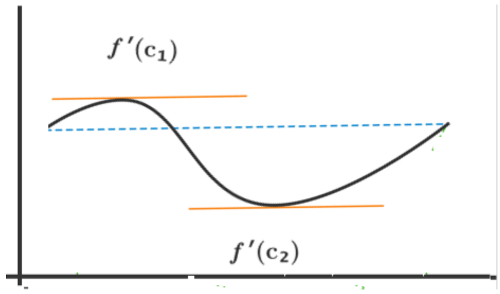
On the interval $[0,5]$, the absolute maximum occurs at $t = 2$, at which time the stone reaches a height of 144 ft. Because $f'(t)$ is the velocity of the stone, the maximum height occurs at the instant the velocity is zero.

Section 4.2 - Mean Value Theorem

The Mean Value Theorem is a cornerstone in the theoretical framework of calculus. Several critical theorems (some stated in previous sections) rely on the Mean Value Theorem; this theorem also appears in practical applications. We begin with a preliminary result known as Rolles Theorem.

Theorem 3 (Rolle's Theorem).
If f be continuous on the closed interval $[a,b]$ and differentiable on (a,b) with $f(a) = f(b)$. Then, there is at least one point c in (a,b) such that

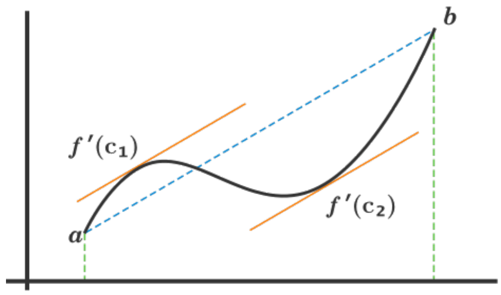
$$f'(c) = 0.$$



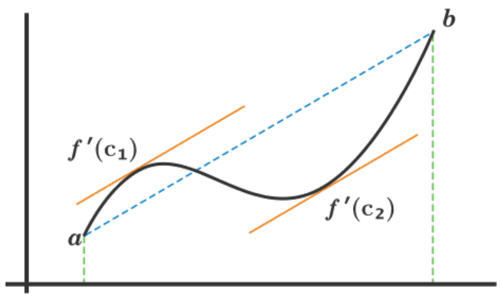
Rolle's Theorem is instrumental in the proof of the Mean Value Theorem. The Mean Value Theorem claims that, for a function f continuos on $[a,b]$ and differentiable on (a,b), there exists a point c in (a,b) at which the slope of the tangent line at c is equal to the slope of the secant line. In other words, we can find a point on the graph of f where the tangent line is parallel to the secant line. More precisely, the theorem is as follows:

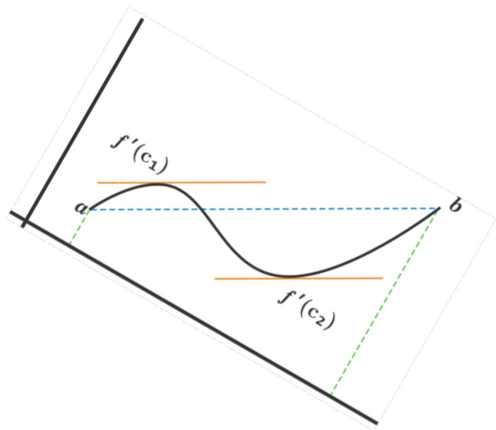
Theorem 4 (Mean Value Theorem).
If f is continuous on the closed interval $[a,b]$ and differentiable on (a,b), then there is at least one point c in (a,b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



Really, Rolle's Theorem and the mean value theorem are pretty much the same – the Mean Value theorem is just Rolle's theorem if you 'tilt your head' a bit. More precisely, look at the pictures above and then look at the one below:





\- Mean Value Theorem Rolle's Theorem

We now state several results that follow from the Mean Value Theorem.

Theorem 5 (Zero Derivative Implies Constant Function).
If f is differentiable and $f'(x) = 0$ at all points of an interval I, *then f is a constant function on* I.

Theorem 6 (Functions with Equal Derivatives Differ by a Constant).
If two functions have the property that $f'(x) = g'(x)$, for all x of an interval I, *then $f(x) - g(x) = \mathrm{C}$ on* I, *where* C *is a constant. That is, f and g differ by at most a constant.*

Example 5. Find an interval I on which Rolles Theorem applies to $f(x) = x^3 - 7x^2 + 10x$. Then find all points c in I at which $f'(c) = 0$.

Because f is a polynomial, it is everywhere continuous and differentiable. We need an interval $[a, b]$ with the property that $f(a) = f(b)$. Noting that $f(x) = x(x-2)(x-5)$, we choose the interval $[0, 5]$, because $f(0) = f(5) = 0$ (other intervals are possible). The goal is to find points c in the interval $(0, 5)$ at which $f'(c) = 0$, which amounts to the familiar task of finding the critical points of f. The critical points satisfy

$$f'(x) = 3x^2 - 14x + 10 = 0.$$

Using the quadradic formula, the roots are

$$x = \frac{7 \pm \sqrt{19}}{3}, \quad \text{or} \quad x \approx 0.88 \quad \text{and} \quad x \approx 3.79.$$

Hence, the graph of f has two points at which the tangent line is horizontal.

Example 6. Determine whether the function $f(x) = 2x^3 - 3x + 1$ satisfies the conditions of the Mean Value Theorem on the interval $[-2, 2]$. If so, find the point(s) guaranteed to exist by the theorem.

The polynomial f is everywhere continuous and differentiable, so it satisfies the conditions of the Mean Value Theorem. The average rate of change of the function on the interval $[-2, 2]$ is

$$\frac{f(2) - f(-2)}{2 - (-2)} = \frac{11 - (-9)}{4} = 5.$$

The goal is to find points in $[-2, 2]$ at which the line tangent to the curve has a slope of 5 – that is, to find points at which $f'(x) = 5$. Differentiating f, this condition becomes

$$f'(x) = 6x^2 - 3 = 5.$$

With some simple algebra, we can see that this is equivalent to $x^2 = 4/3$. Therefore, the points guaranteed to exist by the Mean Value Theorem are $x = \pm 2/\sqrt{3} \approx \pm 1.15$. The tangent lines have slope 5 at the corresponding points on the curve.

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Section 4.3 - What Derivatives Tell Us

In the section 4.1, we saw that the derivative is a tool for finding critical points, which are related to local maxima and minima. As we show in this section, derivatives (first and second derivatives) tell us much more about the behavior of functions.

> **Theorem 7** (Increasing and Decreasing Functions).
> *Suppose a function f is defined on an interval* I. *We say that f is increasing on* I *if $f(x_2) > f(x_1)$ whenever x_1 and x_2 are in* I *and $x_2 > x_1$. We say that f is decreasing on* I *if $f(x_2) < f(x_1)$ whenever x_1 and x_2 are in* I *and $x_2 > x_1$.*

Recall that the derivative of a function gives the slopes of tangent lines. If the derivative is positive on an interval, the tangent lines on that interval have positive slopes, and the function is increasing on the interval. Said differently, positive derivatives on an interval imply positive rates of change on the interval, which, in turn, indicate an increase in function values.

Similarly, if the derivative is negative on an interval, the tangent lines on that interval have negative slopes, and the function is decreasing on that interval.

> **Theorem 8** (Test for Intervals of Increase and Decrease).
> *Suppose f is continuous on an interval* I *and differentiable at all interior points of* I. *If $f'(x) > 0$ at all interior points of* I, *then f is increasing on* I. *If $f'(x) < 0$ at all interior points of* I, *then f is decreasing on* I.

Example 7. Find the intervals on which the following functions are increasing and decreasing:

$$1.\ f(x) = xe^{-x}, \quad 2.\ f(x) = 2x^3 + 3x^2 + 1.$$

1. By the Product Rule, $f'(x) = e^{-x} - xe^{-x} = (1 - x_e^{-x}$. Solving $f'(x) = 0$ and noting that $e^{-x} \neq 0$ for all x, the sole critical point is $x = 1$. Therefore, if f' changes sign, then it does so at $x = 1$ and nowhere else, which implies f' has the same sign throughout each of the intervals $(-\infty, 1)$ and $(1, \infty)$. We determine the sign of f' on each interval by evaluating f' at selected points in each interval:

   - At $x = 0$, $f'(0) = 1 > 0$. So $f' > 0$ on $(-\infty, 1)$, which means that f is increasing on $(-\infty, 1)$.
   - At $x = 2$, $f'(2) = -e^{-2} < 0$. So $f' > 0$ on $(1, \infty)$, which means that f is decreasing on $(1, \infty)$.

   Note also that the graph of f has a horizontal tangent line at $x = 1$.

2. . In this case, $f'(x) = 6x^2 + 6x = 6x(x + 1)$. To find the intervals of increase, we first solve $6x(x + 1) = 0$ and determine that the critical points are $x = 0$ and $x = -1$. If f' changes sign, then it does so at these points and nowhere else; that is, f' has the same sign throughout each of the intervals $(-\infty, -1)$, $(-1, 0)$, and $(0, \infty)$. Evaluating f' at selected points of each interval determines the sign of f' on that interval.

   - At $x = -2$, $f'(-2) = 12 > 0$, so $f' > 0$ and f is increasing on $(-\infty, -1)$.
   - At $x = -1/2$, $f'(-1/2) = -3/2 < 0$, so $f' < 0$ and f is decreasing on $(-1, 0)$.
   - At $x = 1$, $f'(1) = 12 > 0$, so $f' > 0$ and f is increasing on $(0, \infty)$.

   Note also that the graph of f has a horizontal tangent line at $x = -1$ and $x = 0$.

Using what we know about increasing and decreasing functions, we can now identify local extrema.

Theorem 9 (First Derivative Test).
Suppose that f is continuous on an interval that contains a critical point c and assume f is differentiable on an interval containing c, except perhaps at c itself.

- *If f' changes sign from positive to negative as x increases through c, then f has a local maximum at c.*
- *If f' changes sign from negative to positive as x increases through c, then f has a local minimum at c.*
- *If f' is positive on both sides near c or negative on both sides near c, then f has no local extreme value at c.*

Example 8. Consider the function

$$f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1.$$

Find the intervals on which f is increasing and decreasing. Then, find the local extrema of f.

Differentiating f we find that

$$\begin{aligned} f'(x) &= 12x^3 - 12x^2 - 12x + 12 \\ &= 12(x+1)(x-1)^2. \end{aligned}$$

Solving $f'(x) = 0$ gives the critical points $x = -1$ and $x = 1$. The critical points determine the intervals $(-\infty, -1)$, $(-1, 1)$, and $(1, \infty)$ on which f' does not change sign. Choosing a test point in the interval $(-\infty, -1)$, we find that $f' < 0$ and, correspondingly, f is increasing on $(-\infty, -1)$. Similarly, testing points in both $(-1, 1)$ and $(1, \infty)$ we find $f' > 0$ and, correspondingly, f is increasing on both $(-1, 1)$ and $(1, \infty)$.

Now, note that f is a polynomial, so it is continuous on $(-\infty, \infty)$. Because f' changes sign from negative to positive as x passes through the critical point $x = -1$, it follows by the First Derivative Test that f has a local minimum value of $f(-1) = -10$ at $x = -1$. Observe that f' is positive on both sides near $x = 1$, so f does not have a local extreme value at $x = 1$.

Applying Theorem 8 to f' leads to a test for concavity in terms of the second derivative. As such, we now have a useful interpretation of the second derivative: It measures concavity.

Theorem 10 (One Local Extremum Implies Absolute Extremum).
Suppose f is continuous on an interval I *that contains exactly one local extremum at c.*

- *If a local maximum occurs at c, then $f(c)$ is the absolute maximum of f on* I.
- *If a local minimum occurs at c, then $f(c)$ is the absolute minimum of f on* I.

Example 9. Verify that $f(x) = x^x$ has an absolute extreme value on its domain.

First note that f is continuous on its domain $(0,\infty)$. Because $f(x) = x^x = e^{xln(x)}$, it follows that

$$f'(x) = e^{xln(x)}\Big(1 + ln(x)\Big) = x^x\Big(1 + ln(x)\Big).$$

Solving $f'(x) = 0$ gives a single critical point $x = e^{-1}$; there is no point in the domain at which $f'(x)$ does not exist. The critical point splits the domain of f into the intervals $(0,e^{-1})$ and (e^{-1},∞). Evaluating the sign of f' on each interval gives $f'(x) < 0$ on $(0,e^{-1})$ and $f'(x) > 0$ on (e^{-1},∞); therefore, by the First Derivative Test, a local minimum occurs at $x = e^{-1}$. Because it is the only local extremum on $(0,\infty)$, it follows from Theorem 10 that the absolute minimum of f occurs at $x = e^{-1}$. Its value is $f(e^{-1}) \approx 0.69$.

Just as the first derivative is related to the slope of tangent lines, the second derivative also has geometric meaning.

Definition 4 (Concavity and Inflection Points).
Let f be differentiable on an open interval I. If f' is increasing on I, then f is concave upward on I. If f' is decreasing on I, then f is concave downward on I. If f is continuous at c and f changes concavity at c (from up to down, or vice versa), then f has an inflection point at c.

Theorem 11 (Test for Concavity).
Suppose that f'' exists on an open interval I.

- *If $f'' > 0$ on* I, *then f is concave up on* I.
- *If $f'' < 0$ on* I, *then f is concave down on* I.
- *If c is a point of* I *at which f'' changes sign at c (from positive to negative, or vice versa), then f has an inflection point at c.*

Example 10. Identify the intervals on which the function $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1$ is concave up or concave down. Then, locate the inflection points.

This function was considered in Example 6, where we found that $f'(x) = 12(x+1)(x-1)^2$. It follows that $f''(x) = 12(x-1)(3x+1)$. We see that $f''(x) = 0$ at $x = 1$ and $x = -1/3$. These points are candidates for inflection points; to be certain that they are inflection points, we must determine whether the concavity changes at these points. Testing points, we find that $f''(x) > 0$ and f is concave up on $(-\infty,-1/3)$ and $(1,\infty)$. Similarly, $f''(x) < 0$ and f is concave down on $(-1/3,1)$. We see that the sign of f'' changes at $x = -1/3$ and at $x = 1$, so the concavity of f also changes at these points. Therefore, inflection points occur at $x = -1/3$ and $x = 1$.

It is now a short step to a test that uses the second derivative to identify local maxima and minima.

Theorem 12 (Second Derivative Test for Local Extrema).
Suppose that f'' is continuous on an open interval containing c with $f'(c) = 0$.

- *If $f''(c) > 0$, then f has a local minimum at c.*
- *If $f''(c) < 0$, then f has a local maximum at c.*
- *If $f''(c) = 0$, then the test is inconclusive; f may have a local maximum, local minimum, or neither at c.*

Example 11. Use the Second Derivative Test to locate the local extrema of the function $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1$ on $[-2, 2]$.

This function was considered in Examples 6 and 8, where we found that $f'(x) = 12(x+1)(x-1)^2$ and $f''(x) = 12(x-1)(3x+1)$. Therefore, the critical points of f are $x = -1$ and $x = 1$. Evaluating f'' at the critical points, we find that $f''(-1) = 48 > 0$. By the Second Derivative Test, f has a local minimum at $x = -1$. At the other critical point, $f''(1) = 0$, so the test is inconclusive. However, you can check that the first derivative does not change sign at $x = 1$, which means f does not have a local maximum or minimum at $x = 1$.

SECTION 4.4 - GRAPHING FUNCTIONS

We have now collected the tools required for a comprehensive approach to graphing functions. Due the the widespread availability of powerful graphing utilities, graphing is not likely to be a skill that you will explicitly require with any regularity throughout your life. Nonetheless, *understanding* how to graph functions, as well as developing familiarity with the general shape of functions and the information encoded therein provides a powerful and useful sense of *intuition* which is frequently extremely useful when doing mathematics.

The following set of guidelines need not be followed exactly for every function, and you will find that several steps can often be done at once. Depending on the specific problem, some of the steps are best done analytically, while other steps can be done with a graphing utility. Experiment with both approaches and try to find a good balance.

Graphing Guidelines for $\mathbf{y} = \mathbf{f}(\mathbf{x})$

1. . Identify the domain or interval of interest. On what interval(s) should the function be graphed? It may be the domain of the function or some subset of the domain.
2. Exploit symmetry. Take advantage of symmetry. For example, is the function even (i.e. $f(-x) = f(x)$), odd (i.e. $f(-x) = -f(x)$, or neither?
3. Find the first and second derivatives. They are needed to determine extreme values, concavity, inflection points, and intervals of increase and decrease. Computing derivatives – particularly second derivatives – may not be practical, so some functions may need to be graphed without complete derivative information.
4. Find critical points and possible inflection points. Determine points at which $f'(x) = 0$ or f' is undefined. Determine points at which $f''(x) = 0$ or $f^{\{''}$ is undefined.
5. Find intervals on which the function is increasing/decreasing and concave up/down. The first derivative determines the intervals of increase and decrease. The second derivative determines the intervals on which the function is concave up or concave down.
6. Identify extreme values and inflection points. Use either the First or Second Derivative Test to classify the critical points. Both x-coordinates and y-coordinates of maxima, minima, and inflection points are needed for graphing.
7. Locate all asymptotes and determine end behavior. Vertical asymptotes often occur at zeros of denominators. Horizontal asymptotes require examining limits as $x \to \pm\infty$; these limits determine end behavior.
8. Find the intercepts. The y-intercept of the graph is found by setting $x = 0$. The x-intercepts are found by setting $y = 0$; they are the real zeros (or roots) of f (those values of x that satisfy $f(x) = 0$).
9. Choose an appropriate graphing window and plot a graph. Use the results of the previous steps to graph the function. If you use graphing software, check for consistency with your analytic work. Is your graph complete? That is, does it show all the essential details of the function?

Example 12.
Given the following information about the first and second derivatives of a function f that is continuous on $(-\infty,\infty)$, sketch a possible graph of f.

$$f' < 0,\ f'' > 0 \text{ on } (-\infty,0) \qquad f' > 0,\ f'' > 0 \text{ on } (0,1) \qquad f' > 0,\ f'' < 0 \text{ on } (1,2)$$
$$f' < 0,\ f'' < 0 \text{ on } (2,3) \qquad f' < 0,\ f'' > 0 \text{ on } (3,4) \qquad f' > 0,\ f'' > 0 \text{ on } (4,\infty)$$

First, we will use the information above to determine the behavior of f and its graph. For example, on the interval $(-\infty,0)$, f is decreasing and concave up; so we sketch a segment of a curve with these properties on this interval. Continuing in this manner, we obtain a useful summary of the properties of f (Fig. 1). Following this, we can use the information we've summarized to sketch a possible graph of f (Fig. 2). Notice that derivative information is not sufficient to determine the y-coordinates of points on the curve given in the graph.

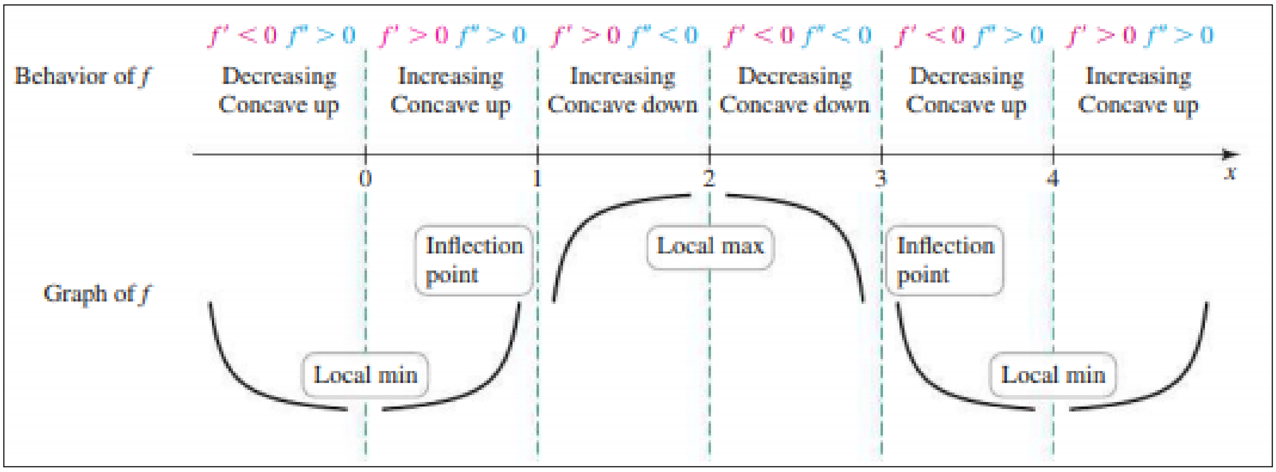


Figure 1: A Summary of the Properties of f

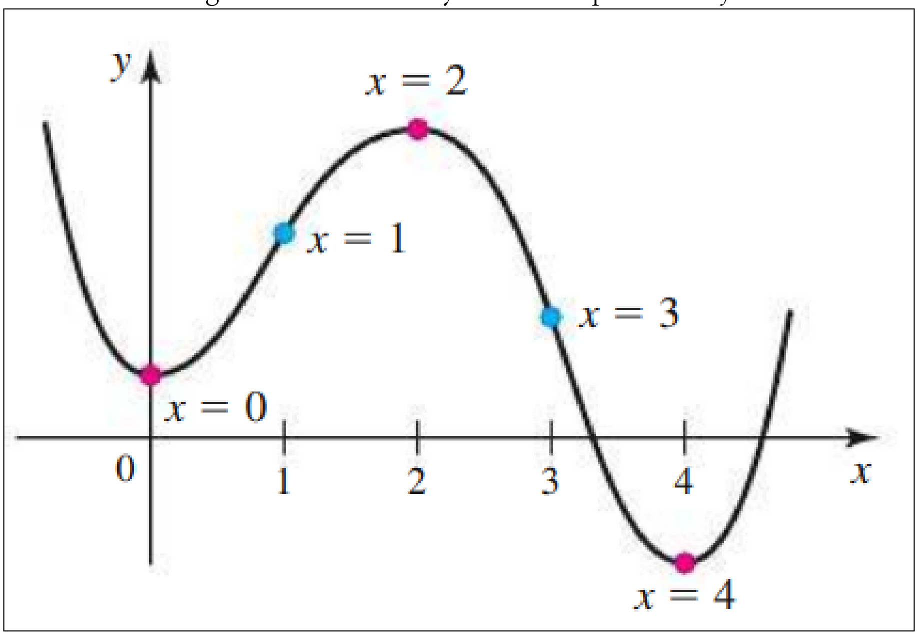


Figure 2: A Graph of f

Section 4.5 - Optimization Problems

The theme of this section is optimization, a topic arising in many disciplines that rely on mathematics. A structural engineer may seek the dimensions of a beam that maximize strength for a specified cost. A packaging designer may seek the dimensions of a container that maximize the volume of the container for a given surface area. Airline strategists need to find the best allocation of airliners among several hubs to minimize fuel costs and maximize passenger miles. In all these examples, the challenge is to find an efficient way to carry out a task, where efficient could mean least expensive, most profitable, least time consuming, or, as you will see, many other measures.

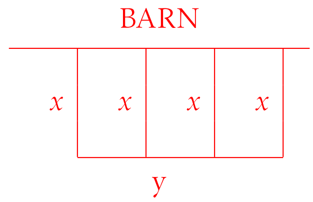
When dealing with optimization problems, we generally wish to optimize a given quantity in some sense under certain conditions imposed on the possible solutions. These conditions imposed on our solutions from the outset of a given problem are called the constraints, while the quantity that we wish to maximize (or minimize in other cases) is called the objective function. At their heart, optimization problems take the following form: What is the maximum (minimum) value of an objective function subject to the given constraint(s)?

Guidelines for Optimization Problems

1. Read the problem carefully, identify the variables, and organize the given information with a picture.
2. Identify the objective function (the function to be optimized). Write it in terms of the variables of the problem.
3. Identify the constraint(s). Write them in terms of the variables of the problem.
4. Use the constraint(s) to eliminate all but one independent variable of the objective function.
5. With the objective function expressed in terms of a single variable, find the interval of interest for that variable.
6. Use methods of calculus to find the absolute maximum or minimum value of the objective function on the interval of interest. If necessary, check the endpoints.

Example 13. A rancher has 400 ft of fence for constructing a rectangular corral. One side of the corral will be formed by a barn and requires no fence. Three exterior fences and two interior fences partition the corral into three rectangular regions. What are the dimensions of the corral that maximize the enclosed area? What is the area of that corral?

We first sketch the corral



where $x \geq 0$ is the width and $y \geq 0$ is the length of the corral. The amount of fence required is $4x + y$, so the constraint is $4x + y = 400$, or $y = 400 - 4x$. The objective function to be maximized is the area of the corral, $A = xy$. Using $y = 400 - 4x$, we eliminate y and express A as a function of x:

$$A = xy = x(400 - 4x) = 400x - 4x^2.$$

Notice that the width of the corral must be at least $x = 0$, and it cannot exceed $x = 100$ (because 400 ft of fence are available). Therefore, we maximize $A(x) = 400x - 4x^2$, for $0 \leq x \leq 100$. The critical points of the objective function satisfy

$$A'(x) = 400 - 8x = 0,$$

which has the solution $x = 50$. To find the absolute maximum value of A, we check the endpoints of $[0, 100]$ and the critical point $x = 50$. Because $A(0) = A(100) = 0$ and $A(50) = 10,000$, the absolute maximum value of A occurs when $x = 50$. Using the constraint, the optimal length of the corral is $y = 400 - 4(50) = 200$. Therefore, the maximum area of $10,000$ ft^2 is achieved with dimensions $x = 50$ ft and $y = 200$ ft.

Example 14. Suppose an airline policy states that all baggage must be box-shaped with a sum of length, width, and height not exceeding 64 in. What are the dimensions and volume of a square-based box with the greatest volume under these conditions?

We consider square-based box whose length and width are w and whose height is h. By the airline policy, the constraint is $2w + h = 64$. The objective function is the volume, $V = w^2 h$. Either w or h may be eliminated from the objective function; the constraint $h = 64 - 2w$ implies that the volume is

$$V = w^2 h = w^2(64 - 2w) = 64w^2 - 2w^3.$$

The objective function has now been expressed in terms of a single variable. Notice that w is nonnegative and cannot exceed 32, so the domain of V is $0 \leq w \leq 32$. The critical points satisfy

$$V'(w) = 128w - 6w^2 = 2w(64 - 3w) = 0,$$

which has roots $w = 0$ and $w = 64/3$. By the First (or Second) Derivative Test, $w = 64/3$ corresponds to a local maximum. At the endpoints, $V(0) = V(32) = 0$. Therefore, the volume function has an absolute maximum of

$$V(64/3) \approx 9709 \ \text{in}^3.$$

The dimensions of the optimal box are $w = 64/3$ in and $h = 64 - 2w = 64/3$ in, so the optimal box is a cube.

Example 15. An 8-foot-tall fence runs parallel to the side of a house 3 feet away. What is the length of the shortest ladder that clears the fence and reaches the house? Assume that the vertical wall of the house and the horizontal ground have infinite extent.

Lets first ask why we expect a minimum ladder length. You could put the foot of the ladder far from the fence, so it clears the fence at a shallow angle; but the ladder would be long. Or you could put the foot of the ladder close to the fence, so it clears the fence at a steep angle; but again, the ladder would be long. Somewhere between these extremes, there is a ladder position that minimizes the ladder length. The objective function in this problem is the ladder length L. The position of the ladder is specified by x, the distance between the foot of the ladder and the fence. The goal is to express L as a function of x, where $x > 0$. The Pythagorean theorem gives the relationship

$$\mathrm{L}^2 = (x+3)^2 + b^2,$$

where b is the height of the top of the ladder above the ground. Similar triangles give the constraint $8/x = b/(3+x)$. We now solve the constraint equation for b and substitute to express L^2 in terms of x:

$$\mathrm{L}^2 = (x+3)^2 + \left(\frac{8(x+3)}{x}\right)^2 = (x+3)^2\left(1 + \frac{64}{x^2}\right).$$

At this juncture, we could find the critical points of L by first solving the preceding equation for L and then solving $\mathrm{L}' = 0$. However, the solution is simplified considerably if we note that L is a nonnegative function. Therefore, L and L^2 have local extrema at the same points, so we choose to minimize L^2. The derivative of L^2 is

$$\frac{d\mathrm{L}^2}{dx} = \frac{d}{dx}\left[(x+3)^2\left(1 + \frac{64}{x^2}\right)\right] = \frac{2(x+3)(x^3-192)}{x^3}.$$

Because $x > 0$, we have $x + 3 \neq 0$; therefore, the condition $\dfrac{d\mathrm{L}^2}{dx} = 0$ becomes $x^3 - 192 = 0$, or $x = 4\sqrt[3]{3} \approx 5.77$. By the First Derivative Test, this critical point corresponds to a local minimum. By Theorem 4.5, this solitary local minimum is also the absolute minimum on the interval $(0, \infty)$. Therefore, the minimum ladder length occurs when the foot of the ladder is approximately 5.77 ft from the fence. We find that $\mathrm{L}^2(5.77) \approx 224.77$ and the minimum ladder length is approximately $\sqrt{224.77} \approx 15$ ft.

SECTION 4.6 - LINEAR APPROXIMATION AND DIFFERENTIALS

Imagine plotting a smooth curve with a graphing utility. Now pick a point P on the curve, draw the line tangent to the curve at P, and zoom in on it several times. As you successively enlarge the curve near P, it looks more and more like the tangent line. This fundamental observation – that smooth curves appear straighter on smaller scales – is called local linearity; it is the basis of many important mathematical ideas, one of which is linear approximation.

Definition 5 (Linear Approximation to f at a).
Suppose f is differentiable on an interval I containing the point a. The linear approximation to f at a is the linear function

$$L(x) = f(a) + f'(a)(x-a), \quad \text{for } x \text{ in } I.$$

Example 16. Suppose you are driving along a highway at a nearly constant speed and you record the number of seconds it takes to travel between two consecutive mile markers. If it takes 60 seconds, to travel one mile, then your average speed is 1 $mi/60s$ or 60 mi/hr. Now suppose that you travel one mile in $60+x$ seconds; for example, if it takes 62 seconds, then $x = 2$, and if it takes 57 seconds, then $x = -3$. In this case, your average speed over one mile is 1 $mi/(60+x)s$. Because there are 3600 s in 1 hr, the function

$$s(x) = \frac{3600}{60+x} = 3600(60+x)^{-1}$$

gives your average speed in mi/hr if you travel one mile in x seconds more or less than 60 seconds. For example, if you travel one mile in 62 seconds, then $x = 2$ and your average speed is $s(2) \approx 58.06$ mi/hr. If you travel one mile in 57 seconds, then $x = -3$ and your average speed is $s(-3) \approx 63.16$ mi/hr. Because you dont want to use a calculator while driving, you need an easy approximation to this function. Use linear approximation to derive such a formula.

The idea is to find the linear approximation to s at the point 0. We first use the Chain Rule to compute

$$s'(x) = -3600(60+x)^{-2},$$

and then note that $s(0) = 60$ and $s'(0) = -3600(60)^{-2} = -1$. Using the linear approximation formula, we find that

$$s(x) \approx L(x) = s(0) + s'(0)(x-0) = 60 - x.$$

For example, if you travel one mile in 62 seconds, then $x = 2$ and your average speed is approximately $L(2) = 58$ mi/hr, which is very close to the exact value given previously. If you travel one mile in 57 seconds, then $x = -3$ and your average speed is approximately $L(-3) = 63$ mi/hr, which again is close to the exact value.

Example 17.
Find the linear approximation to $f(x) = sin(x)$ at $x = 0$ and use it to approximate $sin(\pi/72)$.

We first construct a linear approximation $L(x) = f(a) + f'(a)(x - a)$, where $f(x) = sin(x)$ and $a = 0$. Noting that $f(0) = 0$ and $f'(0) = cos(0) = 1$, we have

$$L(x) = 0 + 1(x - 0) = x.$$

So, our linear approximation yields the estimate

$$sin(\pi/72) \approx L(\pi/72) = \pi/72 \approx 0.04363.$$

A calculator gives $sin(\pi/72) \approx 0.04362$, so the approximation is accurate to four decimal places.

Additional insight into linear approximation is gained by bringing concavity into the picture. Namely, when f is concave up on an interval containing a, the graph of L lies below the graph of f near a. As a result, the linear approximation evaluated at a point near a is *less* than the exact value of f at that point. In other words, the linear approximation *underestimates* values of f near a. To contrast, when f is concave down on an interval containing a, the graph of L lies above the graph of f near a. Correspondingly, this means the linear approximation *overestimates* values of f near a.

We can make another observation related to the degree of concavity (also called curvature). A large value of $|f''(a)|$ (large curvature) means that near $(a, f(a))$, the slope of the curve changes rapidly and the graph of f separates quickly from the tangent line. A small value of $|f''(a)|$ (small curvature) means the slope of the curve changes slowly and the curve is relatively flat near $(a, f(a))$; therefore, the curve remains close to the tangent line. As a result, absolute errors in linear approximation are larger when $|f''(a)|$ is large.

Example 18. Find the linear approximation to $f(x) = \sqrt[3]{x}$ at $x = 1$ and $x = 27$. Then, approximate $\sqrt[3]{2}$ and $\sqrt[3]{26}$. Are these approximations overestimates or underestimates? What is the error on these approximations? Which error is greater? Explain why this is the case.

First, we find the linear approximation to $f(x) = \sqrt[3]{x}$ at $x = 1$ and $x = 27$. Note that $f(1) = 1$ and $f(27) = 3$. Similarly, $f'(x) = (3x)^{-2/3}$, so $f'(1) = 1/3$ and $f'(27) = 1/27$. So, the linear approximation at $x = 1$, denoted L_1, and the linear approximation at $x = 27$, denoted L_2, are given by

$$L_1(x) = 1 + 1/3(x - 1), \quad L_2(x) = 3 + 1/27(x - 27).$$

Using these results, we now find that

$$\sqrt[3]{2} \approx L_1(2) = 4/3 \approx 1.333, \text{ and } \sqrt[3]{26} \approx L_2(26) = 26/27 + 2 \approx 2.963.$$

As $f''(x) = -(2/9)x^{-5/3}$, we see that $f''(x) < 0$ for $x > 0$. Correspondingly, f is concave down at both $x = 1$ and $x = 27$. Hence, both L_1 and L_2 lie above the graph of f and both approximations are overestimates. The error in these approximations are

$$|L_1(2) - \sqrt[3]{2}| \approx 0.073 \text{ and } |L_2(26) - \sqrt[3]{26}| \approx 0.00047.$$

Because $|f''(1)| \approx 0.22$ and $|f''(27)| \approx 0.00091$, the curvature of f is much greater at $x = 1$ than at $x = 27$, explaining why the approximation of $\sqrt[3]{26}$ is more accurate than the approximation of $\sqrt[3]{2}$.

7

SECTION 4.7 - LHÔPITALS RULE

The study of limits in Chapter 2 was thorough but not exhaustive. Some limits, called ndeterminate forms, cannot generally be evaluated using the techniques presented in Chapter 2. These limits tend to be the more interesting limits that arise in practice. A powerful result called LHôpitals Rule enables us to evaluate such limits with relative ease.

Theorem 13 (LHôpitals Rule).
Suppose f and g are differentiable on an open interval I *containing a with $g'(x) \neq 0$ on* I *when $x \neq a$. If $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$, then*

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

provided the limit on the right exists (or is $\pm\infty$). The rule also applies if $x \to a$ is replaced with $x \to \pm\infty$, $x \to a^+$, or $x \to a^-$.

Example 19. Evaluate the limit $\lim_{x\to 1} \dfrac{x^3 + x^2 - 2x}{x - 1}$.

Direct substitution of $x = 1$ into $\dfrac{x^3+x^2-2x}{x-1}$ produces the indeterminate form $\dfrac{0}{0}$. Applying LHôpitals Rule with $f(x) = x^3 + x^2 - 2x$ and $g(x) = x - 1$ gives

$$\lim_{x\to 1} \frac{x^3+x^2-2x}{x-1} = \lim_{x\to 1} \frac{f'(x)}{g'(x)} = \lim_{x\to 1} \frac{3x^2+2x-2}{1} = 3.$$

Example 20. Evaluate the limit $\lim_{x\to 0} \dfrac{\sqrt{9+3x}-3}{x}$.

Substituting $x = 0$ into this function produces the indeterminate form $\dfrac{0}{0}$. Let $f(x) = \sqrt{9+3x} - 3$ and $g(x) = x$, and note that $f'(x) = \dfrac{3}{2\sqrt{9+3x}}$ and $g'(x) = 1$. Applying LHôpitals Rule, we have

$$\lim_{x\to 0} \frac{\sqrt{9+3x}-3}{x} = \lim_{x\to 0} \frac{\frac{3}{2\sqrt{9+3x}}}{1} = \frac{1}{2}.$$

Example 21. Evaluate the limit $\lim_{x\to 0} \frac{e^x - x - 1}{x^2}$.

Substituting $x = 0$ into this function produces the indeterminate form $\frac{0}{0}$. Applying LHôpitals Rule, we have

$$\lim_{x\to 0} \frac{e^x - x - 1}{x^2} = \lim_{x\to 0} \frac{e^x - 1}{2x},$$

which is another limit of the form $\frac{0}{0}$. Therefore, we apply LHôpitals Rule again:

$$\lim_{x\to 0} \frac{e^x - 1}{2x} = \lim_{x\to 0} \frac{e^x}{2} = \frac{1}{2}.$$

Hence, we may conclude that

$$\lim_{x\to 0} \frac{e^x - x - 1}{x^2} = \frac{1}{2}.$$

It is easy to overlook a crucial step in this computation: After applying LHôpitals Rule the first time, you **must** establish that the new limit is an indeterminate form before applying LHôpitals Rule a second time.

LHôpitals Rule also applies directly to limits of the form $\lim_{x\to a} f(x)/g(x)$ where $\lim_{x\to a} f(x) = \pm\infty$ and $\lim_{x\to a} g(x) = \pm\infty$; this indeterminate form is denoted $\frac{\infty}{\infty}$. The proof of this result is found in advanced books.

Theorem 14 (LHôpitals Rule (∞/∞)).
Suppose f and g are differentiable on an open interval I containing a with $g'(x) \neq 0$ on I when $x \neq a$. If $\lim_{x\to a} f(x) = \pm\infty$ and $\lim_{x\to a} g(x) = \pm\infty$, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

provided the limit on the right exists (or is $\pm\infty$). The rule also applies if $x \to a$ is replaced with $x \to \pm\infty$, $x \to a^+$, or $x \to a^-$.

Example 22. Evaluate the limit $\lim_{x\to\infty} \frac{4x^3 - 6x^2 + 1}{2x^3 - 10x + 3}$.

This limit has the indeterminate form ∞/∞ because both the numerator and the denominator approach ∞ as $x \to \infty$. Applying LHôpitals Rule three times, we have

$$\lim_{x\to\infty} \frac{4x^3 - 6x^2 + 1}{2x^3 - 10x + 3} = \lim_{x\to\infty} \frac{12x^2 - 12x}{6x^2 - 10} = \lim_{x\to\infty} \frac{24x - 12}{12x} = \lim_{x\to\infty} \frac{24}{12} = 2.$$

We now consider limits of the form $\lim_{x\to a} f(x)g(x)$, where $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = \pm\infty$. Such limits are denoted $0\cdot\infty$, and LHôpitals Rule cannot be directly applied to limits of this form. Furthermore, its risky to jump to conclusions about such limits. However, as the following example illustrates, this indeterminate form can be recast in the form $0/0$ or ∞/∞ in some cases.

Example 23. Evaluate the limit $\lim_{x\to\infty} x^2 sin\left(\frac{1}{4x^2}\right)$.

This limit has the form $0\cdot\infty$. A common technique that converts this form to either $0/0$ or ∞/∞ is to divide by the reciprocal. We rewrite the limit and apply LHôpitals Rule:

$$\lim_{x\to\infty} x^2 sin\left(\frac{1}{4x^2}\right) = \lim_{x\to\infty} \frac{sin\left(\frac{1}{4x^2}\right)}{\frac{1}{x^2}}.$$

Now, we have a limit of the for $0/0$, and correspondingly may apply LHôpitals Rule:

$$\lim_{x\to\infty} \frac{sin\left(\frac{1}{4x^2}\right)}{\frac{1}{x^2}} = \lim_{x\to\infty} \frac{cos\left(\frac{1}{4x^2}\right)\frac{1}{4}(-2x^{-3})}{(-2x^{-3})} = \frac{1}{4}\lim_{x\to\infty} cos\left(\frac{1}{4x^2}\right) = \frac{1}{4}.$$

Similarly, limits of the form $\lim_{x\to a}[f(x)-g(x)]$, where $\lim_{x\to a} f(x) = \infty$ and $\lim_{x\to a} g(x) = \infty$. Such limits are denoted $\infty-\infty$, and LHôpitals Rule cannot be directly applied to limits of this form. However, as the following example illustrates, this indeterminate form can also be recast in the form $0/0$ or ∞/∞ in some cases.

Example 24. Evaluate the limit $\lim_{x\to\infty}\left[x-\sqrt{x^2-3x}\right]$.

This limit has the form $\infty-\infty$. To recast this limit in a form to which LHôpitals Rule can be applied, we first factor x from the expression and form a quotient as follows:

$$\lim_{x\to\infty}\left[x-\sqrt{x^2-3x}\right] = \lim_{x\to\infty}\left[x-\sqrt{x^2(1-3/x)}\right] = \lim_{x\to\infty} x\left[1-\sqrt{1-3/x}\right] = \lim_{x\to\infty}\frac{1-\sqrt{1-3/x}}{\frac{1}{x}}.$$

This new limit has the form $0/0$, and LHôpitals Rule may be applied. One way to proceed is to use the change of variables $t = 1/x$:

$$\lim_{x\to\infty}\frac{1-\sqrt{1-3/x}}{\frac{1}{x}} = \lim_{t\to 0^+}\frac{1-\sqrt{1-3t}}{t} = \lim_{t\to 0^+}\frac{\frac{3}{2\sqrt{1-3t}}}{1} = \frac{3}{2}.$$

Hence, we may conclude that

$$\lim_{x\to\infty}\left[x-\sqrt{x^2-3x}\right] = \frac{3}{2}.$$

The indeterminate forms 1^∞, 0^0, and ∞^0 all arise in limits of the form $\lim_{x\to a} f(x)^{g(x)}$, where $x \to a$ could also be replaced with $x \to a^\pm$ or $x \to \pm\infty$. LHôpitals Rule cannot be applied directly to these indeterminate forms – the limits must first be expressed in the form $0/0$ or ∞/∞. Here is how we proceed.

The inverse relationship between $ln(x)$ and e^x says that $f^g = e^{g ln(f)}$, so we first write

$$\lim_{x\to a} f(x)^{g(x)} = \lim_{x\to a} e^{g(x) ln\left(f(x)\right)}.$$

By the continuity of the exponential function, we switch the order of the limit and the exponential function (recall, a theorem allowing us to do this was given in chapter 2). Therefore,

$$\lim_{x\to a} f(x)^{g(x)} = \lim_{x\to a} e^{g(x) ln\left(f(x)\right)} = e^{\lim_{x\to a} g(x) ln\left(f(x)\right)},$$

provided $\lim_{x\to a} g(x) ln\left(f(x)\right)$ exists. Therefore, $\lim_{x\to a} f(x)^{g(x)}$ is evaluated using the following two steps:

Theorem 15 (Indeterminate forms 1^∞, 0^0, and ∞^0).
Assume $\lim_{x\to a} f(x)^{g(x)}$ *has the indeterminate form* 1^∞, 0^0, *or* ∞^0.

1. *Analyze* $L = \lim_{x\to a} g(x) ln\left(f(x)\right)$. *This limit can be put in the form* $0/0$ *or* ∞/∞, *both of which are handled by LHôpitals Rule.*
2. *When* L *is finite,* $\lim_{x\to a} f(x)^{g(x)} = e^L$. *If* $L = \infty$ *or* $-\infty$, *then* $\lim_{x\to a} f(x)^{g(x)} = \infty$ *or* $\lim_{x\to a} f(x)^{g(x)} = 0$, *respectively.*

Example 25. Evaluate the limit $\lim_{x\to 0^+} x^x$.

This limit has the form 0^0. Using the given two-step procedure, we note that $x^x = e^{x ln(x)}$ and first evaluate $L = \lim_{x\to 0^+} x ln(x)$. This limit thas the form $0 \cdot \infty$, which may be put in the form ∞/∞ so that LHôpitals Rule can be applied:

$$\lim_{x\to 0^+} x ln(x) = \lim_{x\to 0^+} \frac{ln(x)}{\frac{1}{x}} = \lim_{x\to 0^+} \frac{1/x}{-1/x^2} = \lim_{x\to 0^+} -x = 0.$$

Now, the second step is to exponentiate the limit:

$$\lim_{x\to 0^+} x^x = e^L = e^0 = 1.$$

An important use of LHôpitals Rule is to compare the growth rates of functions.

Definition 6 (Growth Rates of Functions (as $x \to \infty$)).
Suppose f and g are functions with $\lim_{x\to\infty} f(x) = \lim_{x\to\infty} g(x) = \infty$. Then f grows faster than g as $x \to \infty$ if

$$\lim_{x\to\infty} \frac{g(x)}{f(x)} = 0, \quad \text{or, equivalently, if} \quad \lim_{x\to\infty} \frac{f(x)}{g(x)} = \infty.$$

In such cases where f grows faster than g as $x \to \infty$, we denote this by writing $g << f$.

We say the functions f and g have comparable growth rates if

$$\lim_{x\to\infty} \frac{g(x)}{f(x)} = M$$

where $0 < M < \infty$ (i.e. where M is nonzero and finite).

Here's a useful theorem for comparing the rate of growth of some common functions.

Theorem 16 (Ranking Growth Rates as $x \to \infty$).
For positive real numbers p, q, r, and s, and for $b > 1$, we have the following ranking:

$$ln^q(x) << x^p << x^p ln^r(x) << x^{p+s} << b^x << x^x.$$

Example 26. Compare the growth rates as $x \to \infty$ of $f(x) = ln(x)$ and $g(x) = x^p$, where $p > 0$.

The limit of the ratio of the two functions is $\lim_{x\to\infty} \frac{ln(x)}{x^p}$. We may analyze this limit by employing LHôpitals Rule as follows:

$$\lim_{x\to\infty} \frac{ln(x)}{x^p} = \lim_{x\to\infty} \frac{1/x}{px^{p-1}} = \lim_{x\to\infty} \frac{1}{px^p} = 0.$$

As such, we may see that any positive power of x grows faster than $ln(x)$. That is to say, $ln(x) << x^p$, as stated in the above theorem.

Section 4.9 - Antiderivatives

The goal of differentiation is to find the derivative f' of a given function f. The reverse process, called antidifferentiation, is equally important: Given a function f, we look for an antiderivative function F whose derivative is f. That is, a function F such that $F' = f$.

Definition 7 (Antiderivative).
A function F is an antiderivative of f on an interval I provided $F'(x) = f(x)$, for all x in I.

Notice, this function F will not be unique. Namely, as a derivative may be split up over a sum, observe that if $F'(x) = f(x)$, then

$$(F(x) + C)' = f(x)$$

as well for all constants C.

Theorem 17 (Family of Antiderivatives).
Let F *be any antiderivative of* f *on an interval* I. *Then all the antiderivatives of* f *on* I *have the form* $F + C$, *where* C *is an arbitrary constant.*

Example 27. Find all antiderivatives of $f(x) = 3x^2$.

Note that

$$\frac{d}{dx}(x^3) = 3x^2.$$

Therefore, an antiderivative of $f(x) = 3x^2$ is x^3. By Theorem 17, the complete family of antiderivatives is

$$F(x) = x^3 + C,$$

where C is an arbitrary constant.

The notation $\frac{d}{dx}\big(f(x)\big)$ means "take the derivative of $f(x)$ with respect to x". We need analogous notation for antiderivatives. For historical reasons that become apparent in the next chapter, the notation that means "find the antiderivatives of f" is the indefinite integral $\int f(x)\ dx$. Every time an indefinite integral sign $\int$ appears, it is followed by a function called the integrand, which in turn is followed by the differential dx. For what we need to know in this class, dx simply means that x is the independent variable, or the variable of integration. The notation $\int f(x)\ dx$ represents all the antiderivatives of f. When the integrand is a function of a variable different from x – say, g(t) – then we write $\int g(t)\ dt$ to represent the antiderivatives of g For example, in the context of Example 26, we found that

$$\int 3x^2\ dx = x^3 + C.$$

In essence, we may now take all of the derivative formulas presented earlier in the text and write them in terms of indefinite integrals. We begin with the Power Rule:

Theorem 18 (Power Frule for Indefinite Integrals).

$$\textit{For } \ p \neq -1, \ \int x^p \, dx = \frac{1}{p+1} x^{p+1} + C.$$

A variety of similar results follow:

Theorem 19 (Constant Multiple and Sum Rules).

$$\textbf{\textit{Constant Multiple Rule:}} \ \int k f(x) \ dx = k \int f(x) \ dx \ \textit{ for all real numbers } \ k$$

$$\textbf{\textit{Sum Rule:}} \ \int f(x) + g(x) \ dx = \int f(x) \ dx + \int g(x) \ dx$$

Example 28. Determine the indefinite integral $\int \left(3x^5 + 2 - 5\sqrt{x}\right) \ dx$

$$\begin{aligned}
\int \left(3x^5 + 2 - 5\sqrt{x}\right) \ dx &= \int 3x^5 \ dx + \int 2 \ dx - \int 5\sqrt{x} \ dx \\
&= 3\int x^5 \ dx + \int 2 \ dx - 5\int \sqrt{x} \ dx \\
&= 3\left(\frac{1}{6}x^6\right) + 2x - 5\left(\frac{2}{3}x^{3/2}\right) + C \\
&= \frac{1}{2}x^6 + 2x - \frac{10}{3}x^{3/2} + C
\end{aligned}$$

Example 29. Determine the indefinite integral $\int \left(\frac{4x^{19} - 5x^{-8}}{x^2}\right) \ dx$

$$\begin{aligned}
\int \left(\frac{4x^{19} - 5x^{-8}}{x^2}\right) \ dx &= \int \left(4x^{17} - 5x^{-10}\right) \ dx \\
&= \int 4x^{17} \ dx - \int 5x^{-10} \ dx \\
&= 4\int x^{17} \ dx - 5\int x^{-10} \ dx \\
&= 4\left(\frac{1}{18}x^{18}\right) - 5\left(\frac{-1}{9}x^{-9}\right) + C \\
&= \frac{2}{9}x^{18} - \frac{-5}{9}x^{-9} + C
\end{aligned}$$

Example 30. Determine the indefinite integral $\int \left(z^2+1\right)\left(2z-5\right)\ dz$

$$\begin{aligned}\int \left(z^2+1\right)\left(2z-5\right)\ dz &= \int \left(2z^3-5z^2+2z-5\right)\ dz\\ &= \int 2z^3\ dz - \int 5z^2\ dz + \int 2z\ dz - \int 5\ dz\\ &= 2\int z^3\ dz - 5\int z^2\ dz + 2\int z\ dz - \int 5\ dz\\ &= 2\left(\frac{1}{4}z^4\right) - 5\left(\frac{1}{3}z^3\right) + 2\left(\frac{1}{2}z^2\right) - 5z + C\\ &= \frac{1}{2}z^4 - \frac{5}{3}z^3 + z^2 - 5z + C\end{aligned}$$

To save ourselves some time, here are some lists of common integrals:

1. $\frac{d}{dx}(\sin ax) = a\cos ax \quad \Rightarrow \quad \int \cos ax\, dx = \frac{1}{a}\sin ax + C$
2. $\frac{d}{dx}(\cos ax) = -a\sin ax \quad \Rightarrow \quad \int \sin ax\, dx = -\frac{1}{a}\cos ax + C$
3. $\frac{d}{dx}(\tan ax) = a\sec^2 ax \quad \Rightarrow \quad \int \sec^2 ax\, dx = \frac{1}{a}\tan ax + C$
4. $\frac{d}{dx}(\cot ax) = -a\csc^2 ax \quad \Rightarrow \quad \int \csc^2 ax\, dx = -\frac{1}{a}\cot ax + C$
5. $\frac{d}{dx}(\sec ax) = a\sec ax\tan ax \quad \Rightarrow \quad \int \sec ax\tan ax\, dx = \frac{1}{a}\sec ax + C$
6. $\frac{d}{dx}(\csc ax) = -a\csc ax\cot ax \quad \Rightarrow \quad \int \csc ax\cot ax\, dx = -\frac{1}{a}\csc ax + C$

Figure 3: A List of Common Trigonometric Integrals

7. $\frac{d}{dx}(e^{ax}) = ae^{ax} \quad \Rightarrow \quad \int e^{ax}\, dx = \frac{1}{a}e^{ax} + C$
8. $\frac{d}{dx}(b^x) = b^x\ln b \Rightarrow \int b^x\, dx = \frac{1}{\ln b}b^x + C, b > 0, b \neq 1$
9. $\frac{d}{dx}(\ln|x|) = \frac{1}{x} \quad \Rightarrow \quad \int \frac{dx}{x} = \ln|x| + C$
10. $\frac{d}{dx}\left(\sin^{-1}\frac{x}{a}\right) = \frac{1}{\sqrt{a^2-x^2}} \quad \Rightarrow \quad \int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + C$
11. $\frac{d}{dx}\left(\tan^{-1}\frac{x}{a}\right) = \frac{a}{a^2+x^2} \quad \Rightarrow \quad \int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$
12. $\frac{d}{dx}\left(\sec^{-1}\left|\frac{x}{a}\right|\right) = \frac{a}{x\sqrt{x^2-a^2}} \quad \Rightarrow \quad \int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\left|\frac{x}{a}\right| + C, a > 0$

Figure 4: A List of Other Common Integrals

Example 31. Determine the indefinite integral $\int sec^2(3x)\ \ dx$

Using (3) as given in Figure 1, we have

$$\int sec^2(3x)\ \ dx = \frac{1}{3}tan(3x) + C.$$

Example 32. Determine the indefinite integral $\int cos\left(\frac{x}{2}\right)\ \ dx$

Using (1) as given in Figure 1, we have

$$\int cos\left(\frac{x}{2}\right)\ \ dx = \frac{1}{1/2}sin\left(\frac{x}{2}\right) + C = 2sin\left(\frac{x}{2}\right) + C.$$

Example 33. Determine the indefinite integral $\int e^{-10t}\ \ dt$

Using (7) as given in Figure 2, we have

$$\int e^{-10t}\ \ dt = -\frac{1}{10}e^{-10t} + C.$$

Example 34. Determine the indefinite integral $\int \frac{4}{\sqrt{9-x^2}}\ \ dx$

Using (10) as given in Figure 2, we have

$$\int \frac{4}{\sqrt{9-x^2}}\ \ dx = 4\int \frac{dx}{\sqrt{3^2-x^2}}\ \ dx = 4sin^{-1}\left(\frac{x}{3}\right) + C.$$

Example 35. Determine the indefinite integral $\int \frac{dx}{16x^2+1}$

First, we manipulate the integrand until the integral it is of a form given in either Fig. 1 or Fig. 2:

$$\int \frac{dx}{16x^2+1} = \frac{1}{16}\int \frac{dx}{x^2+\frac{1}{16}} = \frac{1}{16}\int \frac{dx}{x^2+\left(\frac{1}{4}\right)^2}.$$

Now, using (11) as given in Figure 2, we have

$$\int \frac{dx}{16x^2+1} = \frac{1}{16}\int \frac{dx}{x^2+\left(\frac{1}{4}\right)^2} = \frac{1}{16}\left(4tan^{-1}(4x)\right) + C = \frac{1}{4}tan^{-1}(4x) + C.$$

Now that we know some basics about integration, we can introduce the concept of a differential equation. While this is not the focus of this class, it certainly is a powerful tool and being aware of its existence is undeniably of value.

An equation involving an unknown function and its derivatives is called a *differential equation*. Here is an example to get us started. Suppose you know that the derivative of a function f satisfies the equation

$$f6'(x) = 2x + 10.$$

To find a function f that satisfies this equation, we note that the solutions are antiderivatives of $2x + 10$, which are $x^2 + 10x + C$, where C is an arbitrary constant. So we have found an infinite number of solutions, all of the form $f(x) = x^2 + 10x + C$.

Now consider a more general differential equation of the form $f'(x) = g(x)$, where g is given and f is unknown. The solution f consists of the antiderivatives of g, which involve an arbitrary constant. In most practical cases, the differential equation is accompanied by an *initial condition* that allows us to *determine* the arbitrary constant. That is, we consider problems of the form

$$\begin{aligned} f'(x) &= g(x), \ \text{ where } \ g \ \text{ is given, and} \\ f(a) &= b, \ \text{ where } \ a \ \text{ and } \ b \ \text{ are given.} \end{aligned}$$

A differential equation coupled with an initial condition is called an *initial value problem*.

Example 36. Solve the initial value problem $f(x) = x^2 - 2x$ with $f(1) = \frac{1}{3}$.

The solution f is an antiderivative of $x^2 - 2x$. Therefore, f is of the form

$$f(x) = \frac{1}{3}x^3 - x^2 + C,$$

for some constant C. That is, we have determined that the solution is a member of a family of functions, all of which differ by a constant. This family of functions, called the *general solution*, defines a family of curves shifted up or down by various choices of C. Using the initial condition $f(1) = \frac{1}{3}$, we must find the *particular function* belonging to the family of functions given by the general solution which satisfies the initial condition. That is, we must find a member of the family of functions given by the general solution whose graph passes through the point $\left(1, \frac{1}{3}\right)$. To do so, we solve for C as follows:

$$\begin{aligned} f(1) &= \frac{1}{3}(1)^3 - (1)^2 + C \\ \frac{1}{3} &= \frac{1}{3} - 1 + C \\ C &= 1. \end{aligned}$$

Hence, the solution to the initial value problem is

$$f(x) = \frac{1}{3}x^3 - x^2 + 1.$$